

In–Course Assessment Examination – 2

CSC 3213(P) – Computer Architecture (Practical)

Applied Mathematics and Computing

Second Semester – Academic Year 2020/2021

Department of Physical Science

University of Vavuniya

Time Allowed: One hour

Date: 02–June–2025

Instructions

- Create a directory on the desktop with your registration number.
- Take screenshots of your outputs and save them inside this directory.
- Each program must start with a comment line containing your registration number.

Q1

Write a program that prints all numbers between 1 and 100 whose square roots are integers.

Sample Output:

The square numbers from 1 to 100 are:

1	4	9	16	25	36
49	64	81	100		

Q2

Write a program to determine the electricity bill based on the number of units consumed.

Billing Rates:

- 0 to 20 units: Rs. 2 per unit
- 21 to 30 units: Rs. 4 per unit
- Above 30 units: Rs. 10 per unit

The program should accept the total number of units as input and display the corresponding total bill amount.

Example:

Let's say the user inputs **35 units** of electricity usage. The calculation will be:

$$\begin{array}{rcl} \text{First 20 units} & : & 20 \times 2 = 40 \\ \text{Next 10 units (21-30)} & : & 10 \times 4 = 40 \\ \text{Remaining 5 units (31-35)} & : & 5 \times 10 = 50 \\ \textbf{Total Bill} & : & 40 + 40 + 50 = 130 \end{array}$$

Sample Output:

Enter the units:35

Total bill balance:130